

Trekking Management Application

Project Report
Modern Application Development I

1. Student Details

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2. Project Overview

A role-based web application developed using Flask, SQLite, Jinja2, HTML and CSS for managing trekking activities, staff and user bookings.

3. Problem Statement

The application replaces manual trekking management with a centralized system for authentication, trek management, staff approval and booking management.

4. Approach

Implemented role-based authentication, SQLAlchemy models, Admin/Staff/User dashboards, trek management and booking workflows.

5. Technologies Used

Python
Flask
SQLite
SQLAlchemy
Jinja2
HTML
CSS

6. Database Tables

Table	Purpose
User	Stores user, staff and admin data
Trek	Stores trek details
Booking	Stores trek bookings

7. Features Implemented

- Authentication
- Admin: Add/Delete Trek, Approve Staff, Assign Staff
- Staff: View Assigned Treks, Update Status and Slots
- User: View Treks, Book Trek, Booking History

8. ER Diagram

User (1) ----< Booking >---- (1) Trek

9. AI / LLM Declaration

AI assistance was used for learning concepts, debugging and understanding Flask. The application was integrated, tested and understood before submission.

10. Conclusion

The project successfully implements a simple trekking management system using Flask and SQLite.